

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458111.75408 +/- 0.00046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1107 +/- 0.0075
Transit depth (Rp/Rs)^2	1.23 +/- 0.17 [%]
Orbital Inclination (inc)	86.8 +/- 1.86
Airmass coefficient 1 (a1)	1.0352 +/- 0.009
Airmass coefficient 2 (a2)	-0.0332 +/- 0.0052
Scatter in the residuals of the lightcurve fit is	0.87 %
Best Comparison Star	#3 - [243, 206]
Optimal Aperture	3.77
Optimal Annulus	4.85
Transit Duration (day)	0.1281 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1107 $\pm$ 0.0075 vs 0.1178 (deviation 0.95 σ)
Duration fit vs published	0.1281 d vs 0.125 d (deviation 1.48 σ)
Mid-time O-C	1572.4 min
Depth SNR	14.8
Residual scatter	0.87 %

Light curve

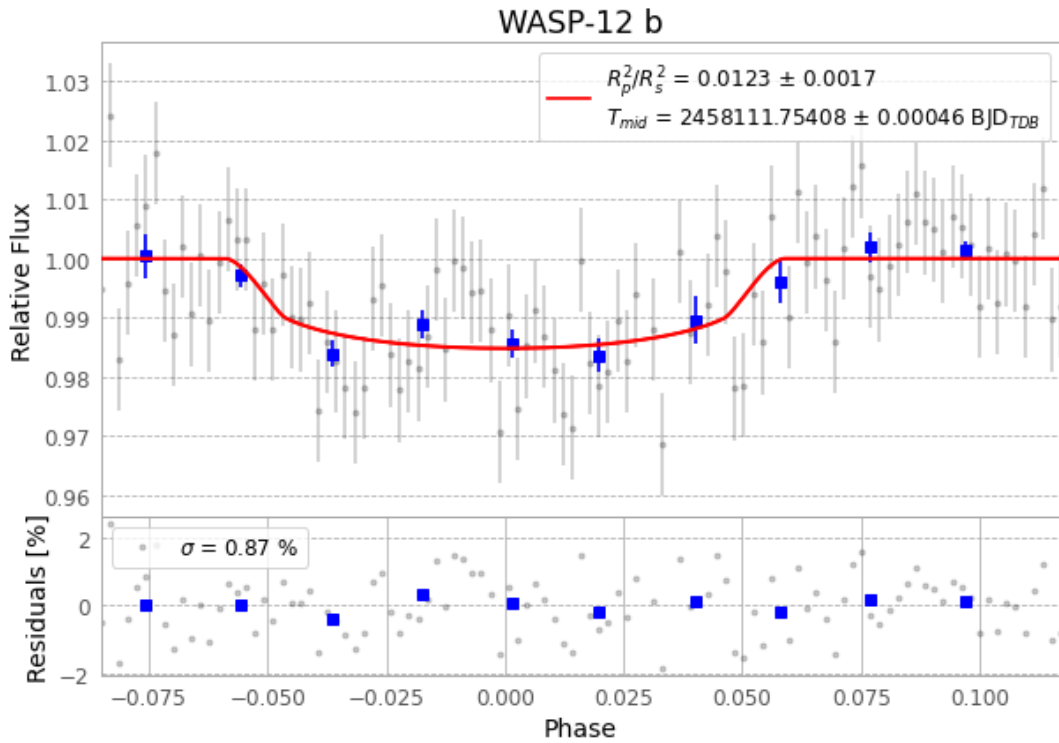


Figure 1 — FinalLightCurve_WASP-12 b_2017-12-23.png (SHA-256 8a689b7750640745...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 56d81193a007d452...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

107 FITS frames (70 MB), WASP-12171224034213.FITS → WASP-12171224090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-171223.log (2732bad54430...), FinalLightCurve_WASP-12 b_2017-12-23.png (8a689b775064...), FinalLightCurve_WASP-12 b_2017-12-23.csv (b89cbb1bbf53...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 15:16Z.